ITA0610-Machine Learning

LAB EXPERIMENTS

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Lab 18: Maximum Likelihood Estimation (MLE)

Aim

To implement **Maximum Likelihood Estimation (MLE)** and estimate the parameters of a normal distribution.

Algorithm

1. Import the required library.
2. Generate sample data from a normal distribution.
3. Calculate the mean of the data.
4. Calculate the variance of the data.
5. Use these values as the MLE estimates.
6. Display the estimated mean and variance.

Program

```
# Lab 18: Maximum Likelihood Estimation


import numpy as np


# Generate sample data

np.random.seed(42)


data = np.random.normal(

   loc=10,

   scale=2,

   size=100

)


# Calculate MLE estimates
```

```python
mu_mle = np.mean(data)


sigma_squared_mle = np.mean(
    (data - mu_mle) ** 2
)


print("MLE Mean:", round(mu_mle, 3))
print("MLE Variance:", round(sigma_squared_mle, 3))
```

Sample Output

MLE Mean: 9.792

MLE Variance: 3.266

Result

Thus, **Maximum Likelihood Estimation was successfully implemented**, and the mean and variance parameters of the normal distribution were estimated from the given data.